

HTU Course No:
10204250

HTU Course Name:
Artificial Intelligence (AI) & Intelligent Systems

BTEC UNIT
No: K/618/5660

BTEC UNIT Name:
Fundamentals of Artificial Intelligence (AI) & Intelligent Systems



Student Name/ID Number/Section	
HTU Course Number and Title	10204250 Artificial Intelligence (AI) & Intelligent Systems
BTEC Unit Number and Title	K/618/5660 Fundamentals of Artificial Intelligence (AI) & Intelligent Systems
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Assignment Author	Rami Alouran
Course Tutor	Rami Alouran
Assignment Title	Designing and Evaluating Intelligent Systems
Assignment Ref No	1
Issue Date	04/08/2025
Formative Assessment dates	From 10/08/2025 to 28/08/2025
Submission Date	07/09/2025
IV Name & Date	Islam Alomari 03/08/2025

Submission Format

Submission Format:

- Technical Report (.docx).
- Intelligent System Implementation (.ipynb).

Reports Guidelines:

- Your submissions should be a soft copy via the eLearning school system. The report should be:
Written in professional style format.
- Include a cover page: Student name, Class, Assignment Title, and Date.
- Your work must be supported with references using the Harvard reference system.
- Plagiarism more than 15%, shall result in failing the course

Oral Exam:

- An oral discussion will be scheduled with your instructors to assess your understanding of the assignment.
- Answering all tasks in the report will be evaluated through the oral assessment. So, if any question in the oral assessment for any criteria is not answered at the required level that means you will lose that criterion.

Unit Learning Outcomes

- LO1** Discuss the theoretical foundation of Artificial Intelligence and its impact on users and organisations.
LO2 Analyse the approaches, techniques and tools to deploy Intelligent Systems in an organisation.
LO3 Modify an AI-based system to improve how exhibits intelligence in response to a real-world problem.
LO4 Evaluate the technical and ethical challenges and opportunities of Intelligent Systems.

Assignment Brief and Guidance

Part 1

Submission date: 13-08-2025 11:59pm

You have been hired as an AI consultant by a mid-sized company seeking to explore the integration of AI into its operations. The company is particularly interested in understanding the technical possibilities of AI in their company, how AI might affect both the organization and its end users, and the associated ethical, legal, and security challenges. Please address the following questions to guide the company's strategic decision making. **Each sentence that includes information derived from an external source should have its own reference to clearly attribute the origin and avoid ambiguity or plagiarism.**

Q1) Explain: A) what Artificial Intelligence is using recent academic or industry sources (published within the last 1-2 years) B) Explain the difference between narrow and general AI C) Define what is Generative AI.

Q2) Select a healthcare organization (e.g., a hospital, clinic, pharmaceutical company, or healthtech startup). Explain why the healthcare sector is particularly suitable for AI applications. What specific challenges or opportunities in this field make it a compelling area for AI integration?

Q3) Based on the organization selected in Q2, answer the following:

A) Identify three possible AI applications the company could use.

B) Use the OpenAI Impact/Effort Matrix (pages 25–27) to rank these applications based on how useful they are and how difficult they would be to implement. Link:

<https://cdn.openai.com/business-guides-and-resources/identifying-and-scaling-ai-use-cases.pdf>

Q4) Identify and explain two ethical or security risks of using AI in your selected organization.

Q5) Discuss two technical challenges in building, deploying, and maintaining the AI systems you proposed building.

Q6) Review possible legal issues related to AI. How might these affect both the user and the company?

Q7) Imagine the company launches an AI system, but later it starts giving wrong or unfair results (e.g., hallucinating, making unfair recommendations). Critically evaluate:

A) What should developers, users, and regulators each do to prevent this kind of problem?

B) Who do you think is most responsible when something goes wrong, and why?

C) How can the company improve the system in the future to reduce these risks?

Part 2

Submission date: 07-09-2025 8:00am

As a newly hired AI consultant at a logistics company operating in Southern California, you have been given access to a historical dataset containing key operational data. The dataset includes fields such as transportation mode, customer region, shipment priority, delivery status, and delivery time. It provides a real-world context for analyzing and improving logistics processes through AI-driven solutions, such as predicting delays, optimizing delivery routes, and enhancing overall supply chain performance. The dataset is available at the link below.

Data source: <https://data.mendeley.com/datasets/8gx2fvg2k6/5>

The primary data file is: DataCoSupplyChainDataset.csv

Q8) This dataset includes two important target variables: Late_delivery_risk and Order_Profit_Per_Order:

A) Identify a suitable machine learning model for each task and justify why each model is appropriate based on the type of prediction and the nature of the data.

B) Describe the main steps involved in building a complete machine learning pipeline (from data preparation to model evaluation).

Q9) You are given two tools to implement your classification model: Python (scikit-learn), and Microsoft Azure ML Studio. Compare the pros and cons of each for: A) Preprocessing data B) Model training and Evaluation

Q10) Demonstrate how you would build two machine learning pipelines in Python one for predicting “late deliveries” and the second to predict “order profit” using the provided dataset. Your pipeline must include the following: A) Exploratory Data Analysis (EDA) B) Feature scaling C) Model training with an appropriate algorithm C) Hyperparameter tuning D) Model evaluation using relevant metrics E) Visualization and interpretation. Submit your well-commented Python notebook showing each step clearly.

Q11) AI systems are often used to predict delays or support decisions in logistics and supply chains, but many users see them as “black boxes”. Why is explainability important for building trust in these systems?

In your answer, include

- A) Two specific types of stakeholders
- B) What could go wrong if they don’t understand how the system works
- C) Your opinion: Should explainability matter more than accuracy in this case? Why?

Q12) Describe how you implemented the baseline k-NN model to predict late deliveries. In your answer, include:

- A) What value of k did you choose, and why?
- B) What distance metric did you use, and why is it suitable?
- C) A code snippet showing your model setup and training

Q13) You have already built a baseline model to predict late deliveries using Python notebooks locally. Now, explore how implementing your model in a cloud-based environment like Azure Machine Learning Studio could help improve its performance, scalability, or deployment efficiency.

- A) What are the expected benefits of moving to Azure compared to your local setup?
- B) What steps did you take to prepare your model for Azure?
- C) Include screenshots of all stages completed within Azure Machine Learning Studio, including data import, preprocessing, model training, evaluation, and deployment.

Q14) Replace your original model (e.g., k-NN **OR** regression) with a simple Artificial Neural Network (ANN). Submit your Python notebook and answer the following:

- A) How did you design your ANN and why did you choose its settings (e.g., layers, activation, optimizer)?
- B) How does its performance compare to your first model? Use at least two metrics (e.g., F1-score, loss) to explain which model performed better.

Learning Outcome	Pass	Merit	Distinction
LO1 Discuss the theoretical foundation of Artificial Intelligence and its impact on users and organisations.	P1 Describe the fundamental aspects of Artificial Intelligence. P2 Describe the types and areas of application to solve current real-world problems.	M1 Analyse the advantages and disadvantages of using Artificial Intelligence to an area of application.	D1 Critique the potential impact of deploying several types, approaches and tools of AI and Intelligent Systems on both users and organisations.
LO2 Analyse the approaches, techniques and tools to deploy Intelligent Systems in an organisation.	P3 Investigate options around the approaches, techniques and tools for the deployment of modern Intelligent Systems. P4 Compare the advantages and challenges of several tools and techniques for the development of Intelligent Systems.	M2 Demonstrate how different approaches and tools work together for the deployment of Intelligent System	D1 Critique the potential impact of deploying several types, approaches and tools of AI and Intelligent Systems on both users and organisations.
LO3 Modify an AI-based system to improve how exhibits intelligence in response to a real-world problem.	P5 Investigate the Technical implementation of an AI-based system. P6 Explore the technical options to enhance the performance on an AI based system. P7 Modify an AI-based system to enhance performance	M3 Demonstrate a Technical modification, to an existing deployment of an AI based system, using benchmarking to enhance its performance.	D2 Evaluate your own role to improve the performance of an AI based system.

LO4 Evaluate the technical and ethical challenges and opportunities of Intelligent Systems	P8 Investigate the security and ethical issues with Intelligent Systems. P9 Discuss the technical challenges involved in managing and maintaining Intelligent Systems.	M4 Review the legal implications and security risks to both users and organisations of using Intelligent Systems	D3 Analyse the technical and ethical challenges while appreciating the opportunities of Intelligent Systems.
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